

Olek Osikowicz

✉ amosikowicz1@sheffield.ac.uk [olek-osikowicz.github.io](https://github.com/olek-osikowicz) [olek-osikowicz](https://github.com/olek-osikowicz) [olek-osikowicz](https://www.linkedin.com/in/olek-osikowicz)

Education

- PhD** **University of Sheffield**, School of Computer Science Sheffield, UK
• Efficient and reliable simulation-based Autonomous Driving Systems testing Sept 2023 – present
• *PhD Supervisors: Donghwan Shin & Phil McMinn*
- BSc** **University of Sheffield**, Computer Science Sheffield, UK
• Graduated with **First-Class Honours** Sept 2020 – June 2023
• Dissertation: *Grounded In Reality Autonomous Driving Systems Testing*

Current Projects

Distributed Data Infrastructure

- Designed and deployed a multi-node computing cluster using Ray and AWS to parallelize software-in-the-loop evaluations
- Significantly improved the throughput of ML evaluation experiments in testing research group

Multi-Fidelity Test Generation

- Designed Multi-Fidelity Bayesian Optimization algorithms for Autonomous Driving Systems (ADS) testing
- Reduced costs of driving models evaluation by 16.8% compared to state-of-the-art baselines

Generative Modelling for Driving Scenarios

- Implemented a Variational Auto-encoder (VAE) model to learn compact latent representations of driving scenarios
- Designed, trained and evaluated the model enabling efficient sampling of driving scenarios for ADS testing

Flaky Test Analysis for ADS

- Empirically discovered and analysed the causes of flaky tests in simulation-based ADS testing
- Reported and published mitigation guidelines to reduce the impact of flaky tests on ADS verification

Employment

- University of Sheffield**, Research Assistant in Simulation-Based Testing Sheffield, UK
• Developing automated Python tooling for large-scale ADS simulation and testing June 2025 – present
• Integrated physics-based simulator CARLA with AutoWare for closed-loop evaluation
- Dover Fueling Solutions**, Summer Intern Kraków, Poland
• Built and validated automated ETL data pipelines on Microsoft Azure June 2022 – Sept 2022
• Worked with SQL warehouses and Databricks for scalable data processing

Skills

Programming: Python (advanced), C++, SQL, TypeScript, React Native, Svelte

Distributed computing: Ray, Docker, AWS (EC2/S3), GCP, SLURM, Multiprocessing, with Loki, Prometheus, Grafana

Machine Learning: PyTorch, Generative Models (VAEs), Bayesian Optimization, Reinforcement Learning

Publications

Multi-Fidelity Bayesian Optimization for Simulation-Based Autonomous Driving Systems Testing Jan 2026

Olek Osikowicz, Phil McMinn, Wei Xing, Donghwan Shin
2026 IEEE Intelligent Vehicles Symposium (IV 2026)

Empirically Evaluating Flaky Tests for Autonomous Driving Systems in Simulated Environments Apr 2025

Olek Osikowicz, Phil McMinn, Donghwan Shin
[10.1109/FTW66604.2025.00009](https://doi.org/10.1109/FTW66604.2025.00009) [🔗](#) *2025 IEEE/ACM International Flaky Tests Workshop (FTW 2025)*